Mega Practice Question Bank: ALT 5 Sketching Graphs

Sandringham College – 8 SEAL Mathematics

Question Booklet

Contents

Practice Questions **2**

Question 1 . . . 2
Question 2 . . . 3
Question 3 . . . 4
Question 4 . . . 5
Question 5 . . . 6
Question 6 . . . 7
Question 7 . . . 8
Question 8 . . . 9
Question 9 . . . 10
Question 10 . . . 11
Question 11 . . . 12
Question 12 . . . 13
Question 13 . . . 14

Practice Questions

Question 1: Sketching Linear Graphs

Question 1: Sketch each of the following lines. Clearly label all axis intercepts with their exact coordinates.

a. $x = 3$

b. $y = -2$

c. $y = 4x$

d. $y = -x$

e. $y = 2x + 4$

f. $y = -3x + 6$

g. $x + y = 5$

h. $2x - y = 8$

i. $x = -\frac{3}{2}$

j. $y = \frac{5}{4}$

k. $y = \frac{2}{3}x$

l. $y = 3x - 2$

m. $y = -2x + 5$

n. $y = \frac{3}{4}x - 1$

o. $3x + 2y = 9$

p. $5x - 3y = 12$

q. $x = \frac{11}{3}$

r. $y = -\frac{7}{5}x$

s. $y = \frac{4}{5}x - 3$

t. $y = -\frac{5}{3}x + \frac{1}{2}$

u. $y = \frac{2}{7}x + \frac{5}{3}$

v. $6x + 7y = 15$

w. $4y - 9x = 17$

x. $-3x + 8y = -11$

Question 2: Midpoint, Distance, and Line Equations

Question 2: For each of the following pairs of points A and B:

(i) Find the coordinates of the midpoint of A and B.

(ii) Find the exact distance between A and B.

(iii) Determine the equation of the line passing through A and B. Write your answer in the form $y = mx + c$.

a. $A(0, 0)$ and $B(6, 8)$

b. $A(1, 2)$ and $B(5, 10)$

c. $A(2, -1)$ and $B(6, 7)$

d. $A(-3, 4)$ and $B(5, -2)$

e. $A(-2, -5)$ and $B(4, 3)$

f. $A(1, 4)$ and $B(4, 8)$

g. $A(-2, 5)$ and $B(3, -1)$

h. $A(-4, -3)$ and $B(1, 6)$

i. $A(3, -2)$ and $B(-1, -7)$

j. $A(5, 0)$ and $B(0, -4)$

k. $A\left(\frac{1}{2}, 2\right)$ and $B\left(\frac{5}{2}, -4\right)$

l. $A\left(-\frac{3}{2}, 1\right)$ and $B\left(\frac{1}{2}, 6\right)$

m. $A\left(\frac{1}{3}, -2\right)$ and $B\left(\frac{4}{3}, 1\right)$

n. $A\left(-2, \frac{1}{2}\right)$ and $B\left(3, -\frac{5}{2}\right)$

o. $A\left(-\frac{5}{4}, \frac{3}{2}\right)$ and $B\left(\frac{3}{4}, -\frac{1}{2}\right)$

Question 3: Line Equation from Gradient and Point

Question 3: A line has a given gradient m and passes through a specific point P. For each of the following, determine the equation of the line. Write your answer in the form $y = mx + c$.

a. $m = 2$, passes through $P(3, 5)$

b. $m = 4$, passes through $P(1, 7)$

c. $m = -3$, passes through $P(2, -4)$

d. $m = -1$, passes through $P(-5, 2)$

e. $m = 5$, passes through $P(-2, -6)$

f. $m = 0$, passes through $P(4, -3)$

g. $m = \frac{1}{2}$, passes through $P(4, 5)$

h. $m = \frac{2}{3}$, passes through $P(6, -1)$

i. $m = -\frac{3}{4}$, passes through $P(8, 2)$

j. $m = \frac{3}{2}$, passes through $P(-4, -5)$

k. $m = -\frac{1}{3}$, passes through $P(-3, 7)$

l. $m = \frac{5}{4}$, passes through $P(2, 3)$

m. $m = \frac{2}{5}$, passes through $P(1, 4)$

n. $m = -\frac{3}{7}$, passes through $P(-2, 1)$

o. $m = \frac{4}{3}$, passes through $P\left(\frac{1}{2}, 2\right)$

p. $m = -\frac{5}{2}$, passes through $P\left(-\frac{3}{2}, \frac{1}{4}\right)$

q. $m = \frac{1}{6}$, passes through $P\left(\frac{2}{3}, -\frac{5}{6}\right)$

r. $m = -\frac{4}{5}$, passes through $P\left(-\frac{1}{4}, -\frac{3}{5}\right)$

Question 4: Perpendicular Line Equations

Question 4: For each line L, determine the equation of the line perpendicular to L which passes through the given point P. Write your final answers in the form $y = mx + c$.

a. $L : y = 2x + 1$, passes through $P(4, 3)$

b. $L : y = -3x - 2$, passes through $P(6, 1)$

c. $L : y = x + 5$, passes through $P(-2, 4)$

d. $L : y = -x + 3$, passes through $P(5, -2)$

e. $L : y = 4x$, passes through $P(8, -1)$

f. $L : y = -5x + 7$, passes through $P(-10, 3)$

g. $L : y = \frac{3}{4}x - 2$, passes through $P(6, 5)$

h. $L : y = -\frac{2}{5}x + 1$, passes through $P(4, -3)$

i. $L : y = \frac{1}{3}x + 4$, passes through $P(2, 7)$

j. $L : y = -\frac{5}{2}x - 6$, passes through $P(-5, 1)$

k. $L : y = \frac{7}{3}x$, passes through $P(14, -2)$

l. $L : y = -\frac{3}{8}x + 5$, passes through $P(-3, -2)$

m. $L : y = \frac{4}{5}x - 1$, passes through $P(3, 2)$

n. $L : y = -\frac{2}{7}x + 3$, passes through $P(-1, 4)$

o. $L : y = \frac{5}{6}x - 2$, passes through $P(2, -3)$

p. $L : y = -\frac{3}{2}x + \frac{1}{2}$, passes through $P\left(\frac{1}{2}, 3\right)$

q. $L : y = \frac{7}{4}x - \frac{3}{4}$, passes through $P\left(-\frac{3}{2}, \frac{5}{2}\right)$

r. $L : y = -\frac{11}{5}x + \frac{2}{3}$, passes through $P\left(\frac{1}{3}, -\frac{1}{4}\right)$

Question 5: Parallel, Perpendicular, or Neither

Question 5: For each of the following pairs of linear equations, determine if the two lines are parallel, perpendicular, or neither. You must show your reasoning by finding and comparing their gradients.

a. $y = 3x + 2$ and $y = 3x - 5$

b. $y = 4x - 1$ and $y = -\frac{1}{4}x + 2$

c. $y = 2x + 7$ and $y = -2x + 1$

d. $y = -5x + 3$ and $y = -5x - 4$

e. $y = \frac{1}{2}x + 6$ and $y = -2x - 3$

f. $y = \frac{3}{4}x + 1$ and $y = \frac{4}{3}x - 2$

g. $y = -x + 8$ and $y = x + 5$

h. $y = 6x$ and $y = 6x + 9$

i. $2y = 8x + 4$ and $y = 4x - 7$

j. $3x + y = 5$ and $y = \frac{1}{3}x + 2$

k. $4x - 2y = 8$ and $y = -2x + 5$

l. $5x + y = 10$ and $10x + 2y = -4$

m. $2x - 3y = 6$ and $3x + 2y = 4$

n. $x - 4y = 8$ and $4y = x - 12$

o. $x + 2y = 6$ and $2x - y = 3$

p. $3x - 5y = 15$ and $5x - 3y = 9$

q. $4y - 3x = 12$ and $3y + 4x = -6$

r. $7x = 2y + 5$ and $14x - 4y = 9$

s. $-5x = 2 - 3y$ and $6y + 10x = 1$

t. $8y + 5x = 10$ and $16y = 10x + 3$

u. $2y - 9x = 4$ and $9y + 2x = 7$

v. $6x = 11 - 5y$ and $15y + 18x = -2$

w. $\frac{1}{2}y - 2x = 3$ and $y = -\frac{1}{4}x + 1$

x. $\frac{2}{3}x + \frac{1}{4}y = 1$ and $8x + 3y = 12$

Question 6: Point of Intersection

Question 6: Find the exact coordinates of the point of intersection between the following pairs of lines by solving the simultaneous equations. Show your method clearly.

a. $x + y = 5$, $x - y = 1$

b. $2x + y = 7$, $x - y = 2$

c. $3x + 2y = 12$, $x + 2y = 8$

d. $y = 2x + 1$, $y = -x + 4$

e. $4x - 3y = 14$, $2x + 3y = 4$

f. $y = 3x - 5$, $2x + y = 10$

g. $5x + 2y = 20$, $5x - y = 5$

h. $x = 3y + 2$, $2x - y = 9$

i. $3x + 4y = 7$, $2x - y = 5$

j. $5x - 2y = 4$, $3x + 5y = 1$

k. $2x + 3y = 8$, $5x - 4y = -2$

l. $4x - 5y = 10$, $3x + 2y = -1$

m. $-2x + 7y = 3$, $4x - 3y = 5$

n. $y = \frac{1}{2}x + 3$, $y = -\frac{3}{2}x - 1$

o. $6x - 5y = 2$, $2x + 3y = 7$

p. $3x + y = -4$, $5x - 2y = 7$

q. $2(x - 1) + 3y = 5$, $4x - (y + 2) = 3$

r. $\frac{x}{2} + \frac{y}{3} = 1$, $\frac{x}{4} - \frac{y}{2} = 2$

s. $5x - 6y = 11$, $7x + 4y = -3$

t. $y = \frac{2}{5}x - 1$, $3x + 4y = 8$

u. $0.5x - 0.2y = 1.4$, $0.3x + 0.5y = -0.4$

v. $8x - 9y = 2$, $5x + 7y = -6$

w. $3y = 4x - 5$, $5y = 2x + 7$

x. $\frac{2x - 1}{3} + y = 4$, $x - \frac{y + 2}{4} = 1$

Question 7: Features of Quadratics

Question 7: For each of the following quadratic functions, find the exact coordinates of the:

(i) x-intercepts
(ii) y-intercept
(iii) turning point

Factorised Form:

a. $y = (x-2)(x-4)$

b. $y = -(x+1)(x-5)$

c. $y = 2(x-3)(x+3)$

d. $y = -3(x+2)(x-4)$

e. $y = 4(x-1)(x+6)$

f. $y = \frac{1}{2}(x+4)(x-8)$

g. $y = -2(2x-1)(x+3)$

h. $y = \frac{2}{3}(3x-2)(x-4)$

Turning Point Form:

d. $y = (x-3)^2 - 4$

e. $y = -(x+2)^2 + 9$

f. $y = 2(x-1)^2 - 8$

g. $y = -3(x+4)^2 + 27$

h. $y = \frac{1}{2}(x-2)^2 - 8$

i. $y = 4(x+1)^2 - 25$

j. $y = -\frac{1}{3}(x-6)^2 + 12$

k. $y = \frac{3}{4}(x+2)^2 - 27$

General Form:

d. $y = x^2 - 6x + 5$

e. $y = x^2 + 4x - 12$

f. $y = -x^2 + 8x - 15$

g. $y = 2x^2 - 4x - 6$

h. $y = x^2 - 4x + 1$

i. $y = -2x^2 + 8x - 3$

j. $y = 3x^2 + 6x - 2$

k. $y = \frac{1}{2}x^2 - 3x + \frac{5}{2}$

Question 8: Quadratic Tables, Sketching & Intersection

Question 8: For each of the following quadratics, complete the three parts:

a. Construct a table of values for integer values of x between the bounds of the domain (inclusive of both bounds to help plot the open end-point).

b. Sketch the graph for the given domain. Label end-points with their coordinates (use an open circle for a $<$ bound).

c. Find the exact values of x such that the given horizontal line crosses the quadratic.

1. $y = x^2 - 4x + 3$ for $-1 \leq x < 5$. Line: $y = 8$
2. $y = -x^2 + 2x + 8$ for $-2 \leq x < 4$. Line: $y = 5$
3. $y = 2x^2 - 8x + 6$ for $0 \leq x < 4$. Line: $y = 6$
4. $y = x^2 + 6x + 5$ for $-5 \leq x < -1$. Line: $y = -3$
5. $y = 2x^2 - 4x + 5$ for $-2 \leq x < 4$. Line: $y = 9$
6. $y = -3x^2 + 6x + 2$ for $-1 \leq x < 3$. Line: $y = -1$
7. $y = x^2 - 5x + 4$ for $0 \leq x < 5$. Line: $y = 2$
8. $y = \frac{1}{2}x^2 - 2x + 1$ for $-2 \leq x < 6$. Line: $y = 3$
9. $y = 3x^2 + 3x - 2$ for $-3 \leq x < 2$. Line: $y = 5$
10. $y = -2x^2 - 5x + 4$ for $-4 \leq x < 1$. Line: $y = -1$
11. $y = 4x^2 - 10x + 7$ for $0 \leq x < 3$. Line: $y = 6$
12. $y = \frac{3}{2}x^2 + 2x - 5$ for $-4 \leq x < 2$. Line: $y = 0$

Question 9: Sketching Parabolas from Features

Question 9: For each of the following quadratics, you are given the key features. Sketch the graph of the parabola. Ensure you label the x-intercepts, y-intercept, and turning point with their exact coordinates.

a. $y = (x-2)(x-6)$
x-ints: $(2,0)$ and $(6,0)$ **y-int:** $(0,12)$ **TP:** $(4,-4)$

b. $y = -(x+1)(x-3)$
x-ints: $(-1,0)$ and $(3,0)$ **y-int:** $(0,3)$ **TP:** $(1,4)$

c. $y = (x+4)(x+2)$
x-ints: $(-4,0)$ and $(-2,0)$ **y-int:** $(0,8)$ **TP:** $(-3,-1)$

d. $y = -(x-1)(x-7)$
x-ints: $(1,0)$ and $(7,0)$ **y-int:** $(0,-7)$ **TP:** $(4,9)$

e. $y = 2(x-1)(x-5)$
x-ints: $(1,0)$ and $(5,0)$ **y-int:** $(0,10)$ **TP:** $(3,-8)$

f. $y = -2(x+3)(x-1)$
x-ints: $(-3,0)$ and $(1,0)$ **y-int:** $(0,6)$ **TP:** $(-1,8)$

g. $y = \frac{1}{2}(x+2)(x-6)$
x-ints: $(-2,0)$ and $(6,0)$ **y-int:** $(0,-6)$ **TP:** $(2,-8)$

h. $y = -\frac{1}{2}(x-2)(x-8)$
x-ints: $(2,0)$ and $(8,0)$ **y-int:** $(0,-8)$ **TP:** $\left(5,\frac{9}{2}\right)$

i. $y = 3(x+1)(x-2)$
x-ints: $(-1,0)$ and $(2,0)$ **y-int:** $(0,-6)$ **TP:** $\left(\frac{1}{2},-\frac{27}{4}\right)$

j. $y = -3(x-2)(x-5)$
x-ints: $(2,0)$ and $(5,0)$ **y-int:** $(0,-30)$ **TP:** $\left(\frac{7}{2},\frac{27}{4}\right)$

k. $y = \frac{3}{4}(x+4)(x-2)$
x-ints: $(-4,0)$ and $(2,0)$ **y-int:** $(0,-6)$ **TP:** $\left(-1,-\frac{27}{4}\right)$

l. $y = -\frac{2}{3}(x+3)(x-4)$
x-ints: $(-3,0)$ and $(4,0)$ **y-int:** $(0,8)$ **TP:** $\left(\frac{1}{2},\frac{49}{6}\right)$

Question 10: Parabola Equations (Turning Point Form)

Question 10: For each of the following, a parabola has a given turning point and passes through another specific point. Determine the equation of the parabola in turning point form, $y = a(x - h)^2 + k$.

a. Turning point at $(1, 2)$ and passes through $(2, 5)$

b. Turning point at $(-2, -3)$ and passes through $(0, 1)$

c. Turning point at $(3, 5)$ and passes through $(5, -3)$

d. Turning point at $(-4, 1)$ and passes through $(-1, 10)$

e. Turning point at $(0, 4)$ and passes through $(3, -14)$

f. Turning point at $(-1, 6)$ and passes through $(1, -6)$

g. Turning point at $(2, 1)$ and passes through $(4, 3)$

h. Turning point at $(-3, -5)$ and passes through $(1, 3)$

i. Turning point at $(5, 3)$ and passes through $(1, -5)$

j. Turning point at $(-3, 0)$ and passes through $(0, -6)$

k. Turning point at $(0, -5)$ and passes through $(4, 3)$

l. Turning point at $(2, 4)$ and passes through $(5, 1)$

m. Turning point at $(1, -2)$ and passes through $(4, 0)$

n. Turning point at $(-2, 4)$ and passes through $(-5, -1)$

o. Turning point at $(3, 7)$ and passes through $(-1, 2)$

p. Turning point at $(-4, -3)$ and passes through $(1, 17)$

q. Turning point at $\left(\frac{1}{2}, 3\right)$ and passes through $\left(\frac{3}{2}, 5\right)$

r. Turning point at $\left(-1, \frac{1}{2}\right)$ and passes through $(2, 5)$

Question 11: Parabola Equations (Factorised Form)

Question 11: For each of the following, a parabola has the given x-intercepts and passes through another specific point. Determine the equation of the parabola in factorised form, $y = a(x - x_1)(x - x_2)$.

a. x-intercepts at $(-2, 0)$ and $(4, 0)$, passes through $(1, -9)$

b. x-intercepts at $(1, 0)$ and $(5, 0)$, passes through $(3, 8)$

c. x-intercepts at $(-4, 0)$ and $(0, 0)$, passes through $(-2, -12)$

d. x-intercepts at $(-3, 0)$ and $(3, 0)$, passes through $(1, 16)$

e. x-intercepts at $(2, 0)$ and $(6, 0)$, passes through $(3, -3)$

f. x-intercepts at $(-5, 0)$ and $(-1, 0)$, passes through $(-2, -9)$

g. x-intercepts at $(-4, 0)$ and $(2, 0)$, passes through $(0, 4)$

h. x-intercepts at $(1, 0)$ and $(7, 0)$, passes through $(4, 6)$

i. x-intercepts at $(-2, 0)$ and $(8, 0)$, passes through $(3, 10)$

j. x-intercepts at $(-6, 0)$ and $(4, 0)$, passes through $(-1, 15)$

k. x-intercepts at $(-3, 0)$ and $(5, 0)$, passes through $(2, -10)$

l. x-intercepts at $(0, 0)$ and $(10, 0)$, passes through $(2, -8)$

m. x-intercepts at $(-1, 0)$ and $(5, 0)$, passes through $(2, -12)$

n. x-intercepts at $(-2, 0)$ and $(3, 0)$, passes through $(1, 12)$

o. x-intercepts at $(-3, 0)$ and $(4, 0)$, passes through $(2, 10)$

p. x-intercepts at $\left(-\frac{1}{2}, 0\right)$ and $(3, 0)$, passes through $(1, -6)$

q. x-intercepts at $(-2, 0)$ and $\left(\frac{3}{2}, 0\right)$, passes through $(1, 6)$

r. x-intercepts at $(-1, 0)$ and $(4, 0)$, passes through $\left(\frac{1}{2}, -7\right)$

Question 12: Point Transformations

Question 12: For each given point, apply the corresponding transformation and state the new exact coordinates.

a. Translate $(4, -2)$ five units right.

b. Translate $(-3, 7)$ two units down.

c. Translate $(-1, -4)$ six units left.

d. Translate $(0, 5)$ eight units up.

e. Reflect $(8, 3)$ in the x-axis.

f. Reflect $(-4, 9)$ in the y-axis.

g. Reflect $(-6, -2)$ in the x-axis.

h. Reflect $(7, -5)$ in the y-axis.

i. Dilate $(2, 5)$ by factor 3 from the x-axis.

j. Dilate $(4, -6)$ by factor 2 from the y-axis.

k. Translate $\left(\frac{3}{2}, 1\right)$ three units left.

l. Translate $\left(-5, -\frac{7}{2}\right)$ four units up.

m. Reflect $\left(\frac{1}{2}, -\frac{3}{4}\right)$ in the y-axis.

n. Dilate $(-12, 8)$ by factor $\frac{1}{4}$ from the y-axis.

o. Dilate $(10, -15)$ by factor $\frac{2}{5}$ from the x-axis.

p. Dilate $(-9, 6)$ by factor $\frac{4}{3}$ from the y-axis.

q. Dilate $(3, -2)$ by factor $\frac{5}{2}$ from the x-axis.

r. Dilate $\left(-\frac{7}{2}, 5\right)$ by factor 4 from the y-axis.

Question 13: Equation Transformations

Question 13: For each equation, apply the corresponding transformation and state the new equation in its simplest form.

a. Translate $y = x^2$ five units up.

b. Translate $y = x^2$ three units down.

c. Translate $y = x^2$ two units right.

d. Translate $y = x^2$ six units left.

e. Reflect $y = x^2$ in the x-axis.

f. Reflect $y = x^2$ in the y-axis.

g. Dilate $y = x^2$ by factor 4 from the x-axis.

h. Dilate $y = x^2$ by factor $\frac{1}{2}$ from the x-axis.

i. Dilate $y = x^2$ by factor 2 from the y-axis.

j. Dilate $y = x^2$ by factor 5 from the y-axis.

k. Dilate $y = x^2$ by factor $\frac{1}{3}$ from the y-axis.

l. Dilate $y = x^2$ by factor $\frac{1}{5}$ from the y-axis.

m. Translate $y = x^2$ $\frac{3}{2}$ units down.

n. Translate $y = x^2$ $\frac{5}{4}$ units right.

o. Dilate $y = x^2$ by factor $\frac{2}{5}$ from the x-axis.

p. Dilate $y = x^2$ by factor $\frac{3}{4}$ from the y-axis.

q. Dilate $y = x^2$ by factor $\frac{5}{2}$ from the y-axis.

r. Dilate $y = x^2$ by factor $\frac{4}{7}$ from the y-axis.